

PROFESSIONAL PROFILE

I combine deep pre-AI systems engineering (5+ years in C/C++, Linux kernel internals, DPDK/VPP packet processing, and hardware) with modern AI tools (OpenAI Codex Desktop, DeepMind AGY CLI). In my projects, AI is never a fragile prompt gimmick; it is an intelligent cognitive brain orchestrated by a robust, deterministic software layer and production bare-metal Linux infrastructure.

CORE COMPETENCIES & TECHNICAL STACK

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| <b>Systems &amp; Low-Level:</b> C, C++20, Linux Kernel, Device Drivers, MIPS, STM32, DPDK, VPP, DPI, conntrack, POSIX, CMake, Make     | <b>Backend &amp; Databases:</b> Python (FastAPI, asyncpg, Pydantic), Node.js (Fastify, Express), PostgreSQL (18+ migrations), Redis, SQLite  |
| <b>Frontend &amp; Mini Apps:</b> React 19, TypeScript, Telegram Mini Apps API, Tailwind CSS, Vite, Framer Motion, SEO, Core Web Vitals | <b>Infrastructure &amp; AI:</b> Bare-Metal Ubuntu, Docker, Docker Compose, Nginx, Caddy, Cloudflare Tunnels, systemd, Codex Desktop, AGY CLI |

SELECTED FEATURED PROJECTS

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Shermos Bot &amp; Mini App</b> — Telegram service & Mini App for manufacturing company in Bishkek                                                                                                                            | FastAPI, PostgreSQL, Redis, React 19       |
| Webhook + Python worker architecture with FastAPI, asyncpg, Pydantic. Distributed Redis queues with per-user mutex locks. 18 PostgreSQL migrations. 3D rendering pipeline. Telegram Mini App in React 19, TypeScript, and Vite. |                                            |
| <b>Floorball KZ</b> — Official digital platform for Kazakhstan Floorball Federation                                                                                                                                             | React, TypeScript, Tailwind CSS, SEO       |
| Engineered with modern frontend standards, sub-second First Contentful Paint, search engine optimization, and seamless responsive behavior across all mobile screen sizes.                                                      |                                            |
| <b>Musikschule CMS Bielefeld</b> — Web platform for music academy in Bielefeld, Germany                                                                                                                                         | React, Vite, Tailwind, Nginx               |
| React + Vite + Tailwind + Framer Motion. Image optimization pipeline with AVIF/WebP and responsive srcSet. Self-hosted Nginx, automated Let's Encrypt (ACME) certificates, HTTP/2, HTTP/3, and edge caching.                    |                                            |
| <b>Audio Transcription &amp; AI Summary Bot</b> — Automated voice & audio transcription bot with AI summarization                                                                                                               | Python, Telegram API, AssemblyAI, Codex AI |
| Processing pipeline integrated with AssemblyAI and OpenAI/Codex APIs. Asynchronous worker process managed via systemd on a private Linux server with resilient job queue handling.                                              |                                            |
| <b>Emily Brautmoden Boutique</b> — Luxury bridal boutique catalog & showcase in Germany                                                                                                                                         | React 19, TypeScript, i18n, Framer Motion  |
| React 19, TypeScript, Vite, modular CSS with design tokens, framer-motion, react-i18next (DE/EN/RU), react-router-dom 7, automated media asset pipeline, Netlify CLI deployment.                                                |                                            |

## PROFESSIONAL WORK EXPERIENCE

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### Software Engineer / Independent Contractor | Freelance & Own Projects

Mar 2025 – Present · Remote (Clients from Germany, Kazakhstan, Kyrgyzstan)

- Infrastructure provisioning: AWS EC2, bare-metal Ubuntu servers, Docker / Docker Compose, Nginx, Caddy, systemd units.
- Networking & Security: automated Let's Encrypt (ACME) certificates, Cloudflare tunnels, HTTP/2, HTTP/3, and API edge protection.
- DevOps workflows: idempotent Bash deployment automation, automated PostgreSQL migrations, monitoring, and production profiles.
- Built production Telegram Mini Apps with FastAPI, asyncpg, Redis distributed locks, and React 19.

### Programming Languages Instructor (C / C++) | Luxkod

Oct 2023 – Aug 2024 · Kazakhstan

- Authored and delivered hands-on C and C++ curricula: pointers, memory management, data structures, algorithms.
- Guided students through compiler error diagnosis, memory leak profiling (Valgrind, GDB), and modular project architecture.

### Software Engineer / Automation Manager | TOO Zavod Armatura Taraz

Jan 2023 – Oct 2023 · Taraz, Kazakhstan

- Engineered a dedicated C++ application for automated extraction and generation of production reports from Excel datasets.
- Created VBA automation pipelines for inventory reconciliation, logistics manifests, and delivery contracts.
- Streamlined end-to-end document workflows for cargo logistics and warehouse distribution.

### C Developer / Systems Engineer | Ideco

Nov 2022 – Dec 2022 · Ekaterinburg

- Developed C systems software leveraging DPDK and VPP (Vector Packet Processing) network stacks.
- Engineered control-plane API extensions and automated unit tests for DPI (Deep Packet Inspection) modules.
- Investigated and resolved bugs in traffic-filtering engines: IPS, DPI, and Linux kernel conntrack.
- Maintained CMake build systems and collaborated in Jira and Git.

### Systems & Embedded Software Engineer | In-Soft LLC

Aug 2021 – Oct 2022 · Novosibirsk

- Patched Linux kernel and developed hardware device drivers to expand wireless device capabilities.
- Authored userspace utilities and Bash scripts for dynamic access point configuration.
- Implemented networking services: DHCP, RADIUS, TFTP, SSH, Docker environments, and config parsing via json-c.

### Laboratory Assistant, 3rd Category | Budker Institute of Nuclear Physics (BINP SB RAS)

Oct 2020 – Aug 2021 · Novosibirsk

- Programmed STM32 microcontrollers in C for real-time digital noise filtering from electret microphones.
- Implemented digital signal processing algorithms interfacing with sensitive analog sensors.

## EDUCATION

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### Novosibirsk State University (NSU)

2018–2021 | Новосибирск

Physical & Technical Informatics (2018–2021) · Strong mathematical & physics foundation